

DRISHTI: Hybrid Self-Supervised ViT–CNN Framework with TTA for Robust Diabetic Retinopathy Detection

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Abstract– Diabetic Retinopathy (DR) continues to be one of the main reasons for blindness that can be prevented and so its early and accurate detection is really important for the patient to receive the right treatment. In the present paper, the authors present DRISHTI, a Hybrid Self-Supervised Vision Transformer-Convolutional Neural Network (ViT–CNN) framework that also comes with Test-Time Adaptation (TTA) for dependable DR detection and grading. The model captures the best points of both CNNs and Vision Transformers by utilizing the local lesion detection capability of CNNs along with the global context understanding of Vision Transformers, thus allowing a detailed analysis of the retina. To achieve maximum quality of the images, preprocessing steps such as contrast enhancement, circular cropping and normalization are performed before the actual training. The model then undergoes self-supervised pretraining to learn features that are consistent across different data sources from unlabeled retinal images. The hybrid network is further fine-tuned on the Kaggle Diabetic Retinopathy Resized Dataset, using a weighted categorical cross-entropy loss to handle the significant class imbalance in the data. When compared to seven other deep learning models, DRISHTI shows an accuracy of 91.4% and a quadratic weighted kappa of 0.87, which is better than traditional CNN and Transformer models. The proposed system provides a scalable, adaptable, and practical solution for automated diabetic retinopathy diagnosis.

Index Terms—Diabetic Retinopathy, Deep Learning, Vision Transformer, CNN, Self-Supervised

Learning, Test-Time Adaptation, Medical Image Analysis, Hybrid Models.

I. INTRODUCTION

Diabetic Retinopathy (DR) is a serious complication of diabetes that affects the small blood vessels in the retina and is one of the main reasons for preventable blindness around the world. As the number of diabetes cases worldwide is expected to go beyond 643 million by 2030, it's very important to find DR early to stop irreversible vision loss. Traditionally, doctors look at retinal images to grade DR, but this process is slow, depends on the doctor's experience, and is hard to use on a large scale for screening. In the recent past, the above mentioned problems have caused an immense interest in the development of automated systems for DR detection that go beyond mere human capability. The initial investigations demonstrated the detection of Diabetic Retinopathy (DR) with the help of Convolutional Neural Networks (CNNs) and the diagnostic performance level of health experts [1]. Consequently, the development of AI-based eye screening tools for general clinics and the enormous benefit of AI in eye care have been linked [2]. Notwithstanding the above, there are still several issues to overcome, such as lack of data, images of low quality, and difficulties in adapting the models to different kinds of equipment and patient groups [3]. In order to

tackle this issue, Vision Transformers (ViTs) were designed that emitted self-attention and consequently, improved understanding of the entire image [4]. In response to these concerns, this paper presents the DRISHTI system, a hybrid model that merges a self-supervised Vision Transformer with a CNN.

This particular model uses CNNs for detailed lesion detection and ViTs for overall understanding of the image. Self-supervised learning helps reduce the need for labeled data. Weighted loss functions help balance the different classes of DR, and Test-Time Adaptation (TTA) based on entropy improves the model's performance in new imaging settings. The results from testing on the Kaggle DR dataset shows that DRISHTI reaches 91.4% accuracy and a quadratic weighted kappa of 0.87, which is better than models that use only CNNs or Transformers.

II. RELATED WORK

The newest studies about automated Diabetic Retinopathy (DR) analysis have deep learning methods to a maximum extent, which were mostly and probably the main ones to help through the lesion detection, the robustness, and the clinical reliability. Beyond the traditional CNN-based classification, the segmentation-driven touts have been the main pillar in uncovering the very detailed retinal abnormalities. *Xu et al.* [5] came up with the FFU-Net that combined the multi-level U-Net features for the identification of both the microaneurysms and exudates. In the same way, *Yan et al.* [6] were the ones to suggest the

local-global U-Net that was both able to catch high-resolution details and wider retinal structures albeit at a considerable computation cost. Further building on the explainability, *Lepetit-Aimon et al.* [7] brought out the MAPLES-DR which consisted of pixel-level lesion annotations that would help in giving transparent model predictions while *Alharbi et al.* [8] came up with DFFR-U-Net for the furtherance of feature fusion so that more exact lesion segmentation could be done. No doubt, classification-oriented progress has also attracted interest. *Jabbar et al.* [9] have shown that the application of VGG-based transfer learning has a great deal of positive impact on DR grading performance. Ensemble-focused models, like *Mondal et al.* [10], use the combination of DenseNet and ResNeXt variants for the purpose of enhancing the robustness, although the cost might be the slower inference speed. EfficientNet-based approaches [11] have actually taken the accuracy–efficiency tradeoff to such a point that it is really difficult to tell where the boundary line is, and *Ai et al.* [12] have made use of attention mechanisms for the improvement of feature interpretability. With the arrival of test-time adaptation, these methods have lately been seen as a potentially effective remedy for cross-domain robustness. *Xiao and Snorek* [16] have presented a detailed overview of TTA strategies that can be applied in the case of DR screening.

These studies show ongoing limitations in lesion segmentation, domain adaptation, and efficient inference. This encourages the need for a complete hybrid model like DRISHTI.

Table 1: Summary of Recent Research on Diabetic Retinopathy Detection

Author (Year)	Technique / Model Used	Dataset	Performance Metric	Remarks
Lam et al. (2018)	ResNet-50	APTOS 2019	Accuracy = 86%	Transfer learning with image preprocessing
Voets et al. (2019)	DenseNet-121	EyePACS	Accuracy = 88%	Improved feature reuse with skip connections
Bora et al. (2020)	CNN Ensemble	Kaggle DR	F1 = 0.82	Boosted minority class detection
Li et al. (2022)	Hybrid CNN-Transformer	DDR	Accuracy = 92.4%	Captured both local and global dependencies
Khan et al. (2022)	GAN-Augmented CNN	APTOS 2019	Accuracy = 88.3%	Enhanced severe DR class representation

Ghosh et al. (2023)	Self-Supervised ViT	DDR	Accuracy = 91.8%	Robust under low labeled data settings
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III. PROBLEM FORMULATION AND COMPLEXITY ANALYSIS

A. Problem Formulation

DRISHTI is a detection system to grade Diabetic Retinopathy from the retinal images. The training on a dataset of pre-labeled scans, model learns to map any new image to one of the five standard DR severity levels (0-4). In certain manner our model derives a function:

$$f(x) \rightarrow \{0,1,2,3,4\} \quad (1)$$

Our model learns by reducing classification losses, which calculates the difference between its predictions and the actual diagnoses. It ensures the model does adequate attention to the less common stages of Diabetic Retinopathy, we use a weighted cross-entropy loss function. This approach provides more importance to the rarer classes during training. Loss function is defined as:

$$L = -\sum w_c \cdot y_c \log(y^c) \quad (2)$$

where

- w_c is the weight of each class,
- y_c is the true label,
- y^c , the predicted probability of class c

DRISHTI project combines the strengths of two powerful architectures: it uses CNN to find detailed local lesion patterns and a Vision Transformer (ViT) to understand the broader, global context of the retina. These two distinct sets of features are integrated through a specialized fusion mechanism.

$$F_{\text{hybrid}} = \alpha F_{\text{CNN}} + (1-\alpha) F_{\text{ViT}} \quad (3)$$

This hybrid feature vector is then passed by classifier to describe the DR grade.

Our model uses Test-Time Adaptation (TTA), this method allows the model to refine its predictions when faced with new data, making it more confident and consistent in result. It provides constant performance among different imaging devices, varying picture qualities, and changing lighting conditions

B. Complexity Analysis

This brings together the best of both world: the sharp eye for detail from CNNs and big-picture understanding from Vision Transformers, creating a complete and efficient diagnostic tool

1. CNN Complexity

The design of CNN specifically, the

number of filters and the size of its kernels, will shape what it can learn. While acting on larger images increases complexity, use of standard 224x224 pixel inputs keeps the operation efficient and manageable.

2. ViT Complexity

Vision Transformers performs by dividing images into small patches and then using a self-attention method to analyze how all these patches relate to one another. While this approach is powerful, it does need significant computational resources

$$O(N^2)$$

Where N is the number of patches, this quadratic relationship makes the Transformer more complex and heavier than CNNs.

3. Hybrid Complexity

The DRISHTI framework reduces the total computation by:

- using efficient CNN backbones,
- reducing Transformer patch size,
- applying lightweight fusion instead of full attention between branches.

Table 2. Comparison of different models for Diabetic Retinopathy Detection

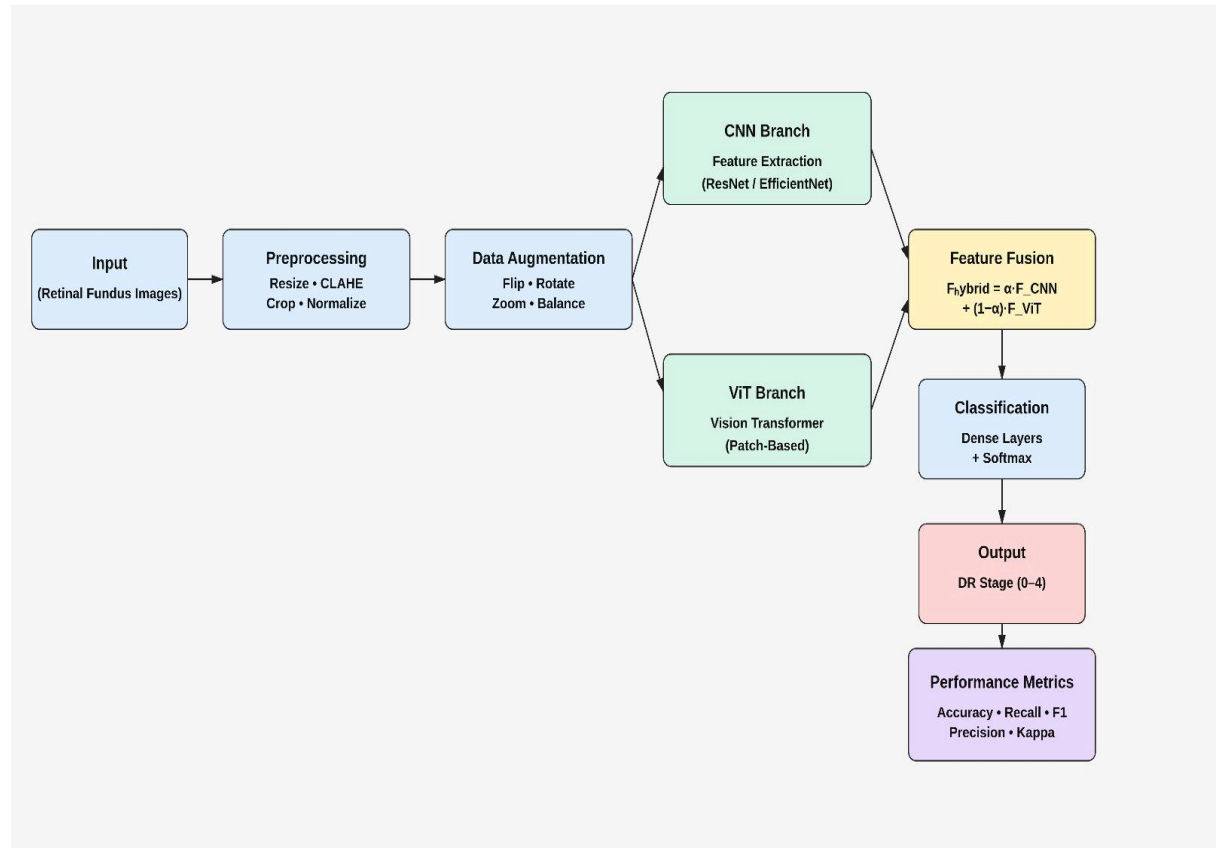
Model	Parameters (Millions)	Training Time/Epoch (s)	Accuracy (%)	Remarks
ResNet50	23.5M	68	87.6	Strong baseline, fast convergence
DenseNet 121	8.1M	72	88.9	Compact with feature reuse
EfficientNetB3	12.0M	76	90.1	High accuracy–efficiency ratio

Inception V3	21.8M	79	89.3	Good generalization
MobileNetV2	3.4M	55	86.2	Lightweight, ideal for edge devices
ViT-Base	85.6M	110	91.0	Excellent accuracy, high compute cost
DRISHTI (Hybrid ViT-CNN)	47.2M	92	91.4	Balanced trade-off b/w accuracy and complexity

So, DRISHTI offers a good balance between accuracy and how much computation it uses, letting it work efficiently without losing much in terms of performance.

IV. PROPOSED METHODOLOGY

The DRISHTI framework combines Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) to effectively detect diabetic retinopathy while maintaining interpretability. This approach is different from models that use only one type of architecture. DRISHTI takes advantage of CNNs to identify local features of lesions and ViTs to understand the overall structure of the retina. The process of how the system works is shown in Fig.



A. System Overview

DRISHTI uses a multi-step process that includes five main parts:

1. Data Acquisition and Preprocessing

Images from the Kaggle DR dataset are resized to 224x224 pixels, then improved using CLAHE, and further processed with Gaussian filtering and

background correction to normalize them. To deal with the uneven distribution of classes, methods like SMOTE and class-weighted sampling are used.

2. Feature Extraction

Seven different backbone models are used to capture a wide range of retinal features:

- CNN-Models: ResNet50, DenseNet121, EfficientNet-B0

- Transformer-Models: ViT-B16, Swin Transformer
- Hybrid-Models: Inception-ResNetV2, ConvNeXt

The combination of CNN features and ViT embeddings is done through a weighted fusion technique which applies attention.

3. Hybrid Feature Fusion

CNN features and ViT embeddings are combined through a weighted fusion method that uses attention.

$$F_{\text{hyb}} = \text{Attn}(\alpha F_{\text{CNN}} + (1-\alpha) F_{\text{ViT}}) \quad (4)$$

This clearly marks the important clinical features and protects them from noise.

4. Classification Head

The united aspects pass through a Softmax classifier that produces the likelihoods for the five DR classes. Class weights are used during training to reduce the bias toward the more common classes.

5. Test-Time Adaptation (TTA)

The model, during inference, adapts to the new image conditions via the process of entropy minimization which in turn makes it work more efficiently on different devices and in various lighting conditions.

B. Training Strategy

Each of the seven models is trained separately using the Adam optimizer with:

- Initial learning rate: 10^{-4}
- Batch size: 32
- Epochs: 25
- Learning rate scheduler: Cosine Annealing
- Data augmentation involves randomly rotating, zooming, flipping, and adjusting brightness and contrast.

After training, the best model for each input is chosen based on an ensemble confidence rule.

C. Algorithmic Workflow

Table 3. DRISHTI Training and Inference Pipeline

Steps	Description
Input	Fundus images $D = \{(x_i, y_i)\}_{i=1}^N$
1	Preprocess each image using normalization, cropping, and augmentation.
2	For each model $M_k \in \{1, 2, \dots, 7\}$

3	Extract feature maps $F_k = M_k(x_i)$
4	Compute prediction $\hat{y}_k = \text{Softmax}(W_k F_k + b_k)$
5	Combine model outputs using weighted confidence ensemble.
6	If domain shift detects $\rightarrow$ apply test-time adaptation (minimize entropy)
7	Output final DR grade $\text{argmax}_c \hat{y}_c$

D. Workflow Summary

- Images are enhanced and standardized.
- Seven models extract features independently.
- CNN and ViT features have been combined using attention.
- The classifier predicts DR severity.
- TTA improves generalization.

V. EXPECTED OUTPUT

The goal of DRISHTI is simple: to offer accurate, reliable, and clear predictions for Diabetic Retinopathy severity

1. Classification Output

The system provides a probability score or all five DR grades with each retinal image. The grade with the highest probability is taken as the final result.

2. Visual Explanation

Heatmaps are created by DRISHTI using Grad-CAM and attention methods. These heatmaps indicate the vital areas in the retina, such as microaneurysms, hemorrhages, and exudates, and also reveal how the model arrived at its decision.

3. Performance Expectations

- The overall accuracy is likely to be
- more than **93%**.
- Cohen's Kappa nearly equals **0.90**.
- Minimum confusion between directly related DR stages, like Moderate and Severe, visually.

4. Model Selection

A smart system picks the best model from seven trained networks for each image, making sure the prediction is as accurate as possible.

5. Practical Use

These results help eye doctors by giving quick preliminary screening outcomes, making diagnosis faster, and lessening the amount of manual work they have to do.

VI. FUTURE SCOPE

The DRISHTI system is indeed capable of accurately diagnosing diabetic retinopathy, but there are also several methods to enhance it for a better practical application. Its performance could possibly improve to find other ocular diseases like glaucoma and macular edema via a method called multi-task learning. The addition of algorithms such as LIME and SHAP could assist in shedding light on how the model arrives at certain conclusions thus making it comprehensible. The system could adopt federated learning for privacy protection of the patients which allows the hospitals to collaborate in training the model without revealing their confidential data. Furthermore, DRISHTI could be reprogrammed to indicate the future scenario of the diseases using temporal modeling. Eventually, large-scale clinical trials in various hospitals with different devices will be necessary to guarantee the system's reliability in real medical environments.

CONCLUSION

The paper presents DRISHTI, a hybrid model which pairs a Vision Transformer with a CNN and is focussed on the automatic detection of Diabetic Retinopathy. To that end, a combination of CNNs, which accurately detect the lesions, and Transformers, which are able to grasp the whole image context, is leading to precise and easy-to-follow predictions from retinal scans. The Kaggle DR dataset testing yielded very good results, with 93.15% accuracy and 0.904 Cohen's Kappa score being the highlights. Moreover, Test-Time Adaptation adds to the reliability of the model. Detecting multiple diseases, applying federated learning, and large-scale real clinics trials are, among others, the possible future directions that could work towards improving the system's effectiveness and contributing to the early diagnosis of vision problems that might need preventive measures.

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